IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2014-15

EEE PART III/IV: MEng, BEng and ACGI

Corrected Copy

OPTOELECTRONICS

Tuesday, 9 December 2:00 pm

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Fundamental constants

$e = 1.6 \times 10^{-19}$ C

$m_e = 9.1 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 = 4\pi \times 10^{-7}$ m kg/C^2

$c = 3 \times 10^8$ m/s

$h = 6.62 \times 10^{-34}$ Js

Maxwell's equations – integral form

$$\int\int_A \underline{\mathbf{D}} \cdot d\underline{a} = \int\int\int_V \rho \, dv$$

$$\int\int_A \underline{\mathbf{B}} \cdot d\underline{a} = 0$$

$$\int_L \underline{\mathbf{E}} \cdot d\underline{L} = -\int\int_A \partial\underline{\mathbf{B}}/\partial t \cdot d\underline{a}$$

$$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \int\int_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$$

Maxwell's equations – differential form

$$\mathrm{div}(\underline{\mathbf{D}}) = \rho$$

$$\mathrm{div}(\underline{\mathbf{B}}) = 0$$

$$\mathrm{curl}(\underline{\mathbf{E}}) = -\partial\underline{\mathbf{B}}/\partial t$$

$$\mathrm{curl}(\underline{\mathbf{H}}) = \underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t$$

Material equations

$$\underline{\mathbf{J}} = \sigma \underline{\mathbf{E}}$$

$$\underline{\mathbf{D}} = \varepsilon \underline{\mathbf{E}}$$

$$\underline{\mathbf{B}} = \mu \underline{\mathbf{H}}$$

Vector calculus (Cartesian co-ordinates)

$$\mathrm{grad}(\phi) = \partial\phi/\partial x \, \underline{i} + \partial\phi/\partial y \, \underline{j} + \partial\phi/\partial z \, \underline{k}$$

$$\mathrm{div}(\underline{\mathbf{F}}) = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$$

$$\mathrm{curl}(\underline{\mathbf{F}}) = \underline{i} \{\partial F_z/\partial y - \partial F_y/\partial z\} + \underline{j} \{\partial F_x/\partial z - \partial F_z/\partial x\} + \underline{k} \{\partial F_y/\partial x - \partial F_x/\partial y\}$$

$$\mathrm{curl}\{\mathrm{curl}(\underline{\mathbf{F}})\} = \mathrm{grad}\{\mathrm{div}(\underline{\mathbf{F}})\} - \nabla^2\underline{\mathbf{F}}$$

$$\int\int_A \underline{\mathbf{F}} \cdot d\underline{a} = \int\int\int_V \mathrm{div}(\underline{\mathbf{F}}) \, dv$$

$$\int_L \underline{\mathbf{F}} \cdot d\underline{L} = \int\int_A \mathrm{curl}(\underline{\mathbf{F}}) \cdot d\underline{a}$$

1. Propagation of TE polarized electromagnetic waves in the (x, z) plane may be described by the scalar wave equation:

$$\partial^2 E_y/\partial x^2 + \partial^2 E_y/\partial z^2 + n^2 k_0^2 E_y = 0$$

Here E_y is the amplitude of the (y-polarised) electric field, n is the refractive index, and $k_0 = 2\pi/\lambda$ where λ is the free-space wavelength.

a) Show that the field $E_y(x, z) = E_0 \exp\{-jnk_0[x \sin(\theta) + z \cos(\theta)]\}$ is a solution. What does it represent? Assuming that $\theta \approx 30^o$, sketch lines of constant phase.

[6]

b) Show that the field $E_y(x, z) = E_0 \exp(\gamma x) \exp(-j\beta z)$ can also be a solution. What does this solution represent? Sketch lines of constant amplitude and constant phase. How realistic is the solution, if its amplitude becomes very large as x rises?

[8]

c) Assuming a modal solution in the form $E_y = E(x) \exp(-j\beta z)$, derive the waveguide equation for z-propagating modes. Show that the transverse field $E(x) = E_0 \exp(\gamma x)$ may be a solution. What other solutions are possible, and when?

[6]

2. a) Sketch the spectral variation of attenuation in silica fibre, and outline the major physical mechanisms contributing to loss. Explain how this variation drives the choice of operating wavelength for long-distance optical fibre communication systems.

[6]

b) Sketch the corresponding spectral variation of refractive index. Explain how this variation drives the choice of wavelength for high bit-rate fibre communications. Is this conclusion in conflict with part a)? If so, what can be done to resolve the situation?

[6]

c) The broadening of a pulse of bandwidth $\Delta\omega$ as it propagates through a distance L is $\Delta T = L\, d(1/v_g)/d\omega\, \Delta\omega$, where $v_g = d\omega/dk$ is the group velocity. Show that this is equivalent to $\Delta T = -L(\lambda_0/c)\, d^2n/d\lambda_0^2\, \Delta\lambda_0$, where n is the refractive index of the medium, λ_0 is the free-space wavelength and $\Delta\lambda_0$ is the equivalent spectral range.

[8]

3. a) Sketch the layout and briefly describe the operation of an electro-optic directional coupler switch, explaining what is meant by the 'bar' and 'cross' states.

[6]

b) Draw the schematic of a 4 x 4 non-blocking switch array and sketch the physical arrangement of a coupler-based array. How many couplers are required in a general N x N non-blocking array?

[4]

c) The coupled mode equations of an asynchronous directional coupler are:

$$dA_1/dz + j\kappa A_2 \exp(-j\Delta\beta z) = 0$$

$$dA_2/dz + j\kappa A_1 \exp(+j\Delta\beta z) = 0$$

Here A_1 and A_2 are the amplitudes of the modes in the two guides, κ is the coupling coefficient and $\Delta\beta$ is the dephasing parameter. Without solving the equations, show that they conserve power.

[5]

d) Assuming that the device is synchronous, solve the equations for the boundary conditions $A_1 = 1$, $A_2 = 0$ on $z = 0$. Show that the solutions also conserve power.

[5]

4. a) Explain the operation of a photodiode. Why is a photodiode more efficient than a photoconductive detector?

[7]

b) Sketch and explain the distribution of electrons near the valence and conduction band edges at room temperature in a semiconductor. What does this distribution imply about the relative likelihood of absorption of photons whose energies are i) only just greater and ii) much greater than the band-gap?

[7]

c) Draw the physical arrangement of a surface entry photodiode. How does the result in b) above affect its performance? Sketch the spectral variation in responsivity you would expect.

[6]

5. a) Explain the difference between direct- and indirect-gap materials. Which type of material is required in optoelectronics?

[6]

b) Explain the difference between spontaneous emission and stimulated emission. Which phenomenon is exploited in LEDs and lasers?

[6]

c) Explain why the external efficiency of an LED is so low. Assuming the refractive index of GaAs is 3.5, estimate the external efficiency of a GaAs LED.

[8]

6. A semiconductor laser designed for operation at 1.5 μm wavelength is based on a waveguide with effective index $n_{eff} = 3.5$ and gain coefficient $g = 2355\ m^{-1}$ when lasing.

a) Determine the conditions for round trip resonance, and show how this leads to separate conditions for phase and gain.

[8]

b) Estimate the length of the laser and the spectral separation $\Delta\lambda$ between adjacent longitudinal modes.

[6]

c) A new laser is fabricated, with a different cavity length, and it is noted that its spectral separation $\Delta\lambda$ has double the value found in b). Estimate the gain coefficient now needed for lasing. Comment on the tradeoff between spectral separation and gain.

[6]